

Evan Jones
U.S. Olympic & Paralympic Committee Practicum
Bi-weekly progress report
Week ending April 6th

Mr. Schwartz,

Since our last correspondence I have made some substantial progress on the event and country level dashboards. The event level dashboard has been updated to allow flexibility in the user choice and method of peak age selection and analysis. Additionally, additional charts such as CDF, violin, and PDF difference have also been added to allow different views of the KDE models. A new section has been added to conduct cross-validation of the KDE models in application. Run-time is a bit longer (16 seconds for largest query on basic laptop) for some queries but loading spinners have also been added to reflect this. Having the cross-validation happen in browser allows the user to customize a model to their personal needs and permits total end user exploration of the KDE analysis. For this reason, I have removed schedule plans to pickle and aggregate cross validation results. This should no longer be necessary as the models are all trained at the edge device according to user parameters so there is no need for a central repository of models and tunings. Such a repository would also be impractical to maintain given the sheer size of combinations of different models the end user can construct. On the country dashboard side of things, I created a tableau choropleth that allows user selection of various parameters around country medal performance over time. This tableau visualization was embedded within the dash application country dashboard. I also created two key subsections of the country dashboard for country pipeline comparison and detailed country pipeline analysis. The first of these sections allows the user selection of two countries, a peak age, and various Olympic games and populates a table that illustrates their respective strengths and weaknesses at aggregate and by event/gender levels. This section is further augmented by a column bar chart which shows country wise counts of pipeline age levels (peak/pre/post). The second section focuses on a single country's pipeline. This includes a line plot of their pipeline strength overtime. Additionally, it has drilldowns in the form of tables that show the top prospects for each event/gender combination. The "pipeline strength" metric referenced above has also undergone some prototyping. At present, the best way it seems for me to construct this metric is by evaluating the potential athletes a country has (competed within 1 year in a given event prior to the Olympic start date to assure activity) and evaluate what their age will be at the start of that given Olympic games. I then tabulate the counts of peak/pre/post age groups by country and event/gender. This age range is initially set by KDE +/- .5 standard deviation from distribution mode range but permits a user to select their own desired age ranges. These bins then undergo a weighting process whereby each age group value counts are multiplied by the cumulative probability (pulled from medaling KDE) for that given age range interval then summed to arrive at an aggregate country pipeline strength which is then transformed into a ranking. As an example, if the area under the medaling KDE curve from pre-peak age range is 15% then a countries pre-peak age counts will be multiplied by .15. We do this for peak and post peak ages as well then sum the results. A similar process is also conducted to arrive at event/gender level pipeline rankings. On the data side of things, I have also augmented the dataset to include the new 2022 Olympic results. I have also discovered a lack of data for all Olympic combination events for the four most recent winter games. I can add this manually, if necessary, but I figured I should let you know in case you had some knowledge on/of this. On the code side of things, this week I spent a bit of time refactoring and cleaning up the codebase to make documentation easier and improve extensibility and organization of the repository. In the coming weeks, I plan to spend some time finishing up the country dashboard, and generally polishing up everything. I also plan to take the last week to write up the final report and create a notebook like representation of the report for the methodology page in the dash application. Let me know if you have any questions and I will be happy to add clarification.

Best regards,

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Evan Jones: United States Olympic & Paralympic Committee Revised Practicum Plan

Week	Week Ending	Planned	Completed	Other Notes
1	January 19 th	- Narrow sport scope to single discipline - Get familiarized with data structure and project goals - Exploratory data analysis with an eye towards data cleaning and formatting	- Alpine Skiing selected - Organized and learned about data structure - Identified droppable columns; Noticed fair amount of missing entries	
2	January 26 th	- Construct some histograms to test basic age to medaling frequency and relationship - Research Alpine 12 unique events; sport governing bodies; scoring and ranking mechanics - Apply some cursory data cleaning and formatting - Identify and correct datatypes for database import	- Created histogram with clear relationship - Extensive notes and annotations taken on specific sport and event details - Dropped columns which were mostly null or near duplicates - Fixed pandas typing issues - Create coding system for missing, blank, nan, and other edge case field entries - Researched methods for filling missing data	
3	February 2 nd	- Create local MySQL database and load in data - Create python Selenium script for datamining International Ski Federation website for athletes - Separate and split team events from solo events - Compare histograms for different gender/event pairs to check similarities (see if sports should be combined or not)	- Local database created and working - Selenium script created and working; filled most missing values for age; scraped profile photos for visualization purposes later. - Data separated by team and gender groupings - Utility of combining event data still unclear further analysis needed	

4	February 9 th	- Organize data into format ready for KDE - Apply KDE to Men's Super G as a test run; Tune bandwidth; Test target variable approach medaling in general vs. specific medal. - Compile list of regulation changes for events - Add documentation to Selenium script/notebook	- Data formatted for KDE; initial results look promising. - Created bandwidth tuning interactive sliders for ad-hoc experimentation and visualization - Compiled list of impactful regulation changes and possible data adjustment remedies - Selenium documentation added	
5	February 16 th	- Research masking methods for train/test given problem setup - Apply remedies for regulations and necessary data adjustments - Fit KDE models on all solo events - Begin work on midterm slide deck	- Train & Testing years/masked determined - Decided to utilize only post reg data, but to include a dashboard view for better understanding of the impact of regulation changes - Fit KDE models on solo events - Developed diagrams for Midterm slide deck -	
6	February 23 rd	- Create country historic and forward-looking pipeline profiles which at each Olympic date contain competitors used; unused; and medal outcomes - Create selenium script for updating country pipelines as new athletes are entered in the FIS database. - Finish midterm slide deck	- Pipeline creation in progress, but not complete as of weekend - Finished midterm report presentation slides - Created flexible support for new athlete entries	Midterm slide deck/report due
7	March 2 nd	- Start layout and design work on Dash dashboard structure - Setup multi-page web app layout - Create landing page - Convert KDE models into dash friendly representation	- Multi-page dash app created - Finalized file structure for the application - Added global scss components and style sheets - Finalized application landing page - KDE models converted to dash interactive plot for customized hyperparameter selection.	Revised Project plan to utilize Dash instead of Tableau causes some changes in overall plan
8	March 9 th	- Create FIS image scraper for athlete profile images - Create all dash components and callbacks for the individual athlete view dashboard page - Create repo for country flags	- Created image scraper used as design elements in dash - Created svg repository for displaying specific country and Olympic flags used as design elements in dash - Finished all callbacks and components on the individual athlete view dashboard	

		- Finish work on KDE descriptive stats and chi squared test	- Began work on Event level dashboard - Took Elliot's advice about viewing age from the individual achievement perspective did some cursory modeling here - Finished KDE analysis	
9	March 16 th	- Finalize individual athlete peak achievement view of age analysis - Add visualization that contrasts the two different age modeling approaches. - Start work on the event view dashboard	- Finished the individual view dashboard - Started work on the new event dashboard - Finished 3 peak age modeling approaches focusing on peak world ranking achieved by event; peak event rank achieved; and peak scaled / smoothed rank achieved. - Visualizations created for both approaches	
10	March 23 ^d	- Finish work on the event view dashboard - Begin country pipeline level analysis and prediction - Begin work on the last country view dashboard	- Significant work accomplished on the event view dashboard, but not complete just yet - Pipeline analysis started and developing well - Created upload page to allow users of the web application the ability to upload new event data via csv which automatically cleans and reconciles the new dataset with the existing dataset. - Methodology page created to layout analysis - Country view dashboard created and work is continuing in development.	
11	March 30 th	- Create the country pipeline comparison dash components - Create the country pipeline detail drill down dash components - Create the country dashboard embedded tableau choropleth -	- Refactored some code and revised the file structure for more extensible usage - Completed the country pipeline comparison dash - Completed the country pipeline detail drill down dash - Added 2022 new Olympic data	
12	April 6 th	- Create a line plot to model and compare country pipeline rankings over time/Olympics - Find a way to incorporate high world ranking athletes into pipeline strength - Create some UI elements for post/pre/peak age gauges	- Completed the country pipeline comparison line plots over time dash component - Completed the Country dashboard UI elements needed - Opted against including world ranking as a weight in pipeline strength the reason being it would be label leakage due to forward looking nature of these events.	

		- Create UI elements for medals awarded	- Created UI elements for gauges and medals.	
13	April 13 th	- Finish country pipeline analysis - Finish country pipeline dash page styling, formatting, text touchup		
14	April 20 th	- Prepare final report and touch up - Organize project files/repo/documentation		Final report due